

Massive Data Fundamentals
(PPOL 5206)

Syllabus

Spring 2026

Summary Information

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OFFICE HOURS: By appointment

CLASS MEETING TIME: Thursdays (6:30 PM – 9:00 PM)

CLASS MEETING LOCATION: Room 502; 125 E St. NW

Course Description

Massive Data Fundamentals equips students to analyze datasets that exceed the capabilities of traditional computing tools. In this hands-on course, students master cloud computing, distributed systems, and the big data technologies essential for understanding modern data challenges in public policy.

Students build practical skills with industry-standard tools including SQL, Apache Spark, Databricks, graph databases, and Snowflake, using Python as the primary programming language. The curriculum progresses through three phases: cloud infrastructure fundamentals, SQL and data engineering, and distributed computing. Students also navigate the evolving landscape of data governance, exploring privacy laws, security protocols, and disaster recovery planning.

The course combines project-based learning with analyses of big data history and discussions of privacy and data governance issues. Students develop end-to-end data capabilities, using cloud platforms to ingest, transform, analyze, and present data, creating portfolio-ready work that demonstrates both technical expertise and policy insight.

Learning Objectives

By semester's end, you will be able to:

Technical Foundations

- Query relational databases using SQL
- Process datasets that exceed single-machine capacity using distributed computing tools
- Navigate cloud platforms to provision and manage scalable data infrastructure
- Use version control and command-line tools that support collaborative data work

Analytical Judgment

- Identify computational, storage, and cost constraints when evaluating data solutions
- Compare architectural approaches (data warehouses, lakes, lakehouses) for policy applications
- Evaluate trade-offs across major cloud platforms for organizational needs

Governance and Context

- Analyze data regulations affecting collection, storage, and cross-border transfer
- Apply ethical frameworks to data-intensive policy decisions

Synthesis

- Design and justify a data architecture recommendation for a policy use case

Grading Policies

Final Grades

The final grade in this course will be determined by two projects, lab results, and participation.

Assignment	Points
Project 1: Global Data Landscape	
Presentation	40
Briefing Report	60
Project 2: Big Data Analytics	
Presentation	40
Website	80
Paper	80
Labs	100
Participation	50
TOTAL	450 points

Project milestones (country selection, outlines, dataset identification, etc.) are graded for completion and are reflected in participation. Their purpose is to provide feedback and identify issues early.

Attendance Policy

Attendance in class is expected, as learning from your peers is an important part of what the school offers. If you are unable to attend, please contact me via email.

Late Assignment Policy

Labs are due by the date and time provided. *Late labs will be penalized 20% for each day overdue.*

Project milestones are completion-based checkpoints: submit something reasonable by the deadline to receive credit. Final project deliverables (presentations, reports, websites, papers) follow the standard late penalty unless you communicate with me in advance about extenuating circumstances.

Participation Policy

Participation in class discussion and on Canvas is encouraged. Please be aware that anything you post online will potentially be public. Classes may be recorded and distributed to students.

Canvas Site

A Canvas course site is set up for this course. Each student is expected to check the site throughout the semester, as Canvas will be the primary venue for outside-classroom communications between the instructor and students. Students can access the course site at <http://canvas.georgetown.edu/>. Support for Canvas is available at (833) 476-1171 or canvas-help@georgetown.edu.

Generative AI Policy

Course AI Policy

Artificial intelligence tools have long been a part of the data science landscape. The introduction of Large Language Models has popularized the use of generative AI, and new applications are being developed every day. I expect that you will use these tools in your future work.

Using machine learning in this course is expected. Generative Business Intelligence tools may be used as part of the learning process. Generative AI may be used to supplement, but not replace, the learning process. Students should document how they used generative AI or other tools as part of a statement when submitting work. I will cover responsible, ethical, and skillful use of AI throughout the course. Wharton's Generative AI Lab provides many useful resources, including tested prompts that can be used for learning: [Wharton Generative AI Labs](https://gail.wharton.upenn.edu/) (<https://gail.wharton.upenn.edu/>)

Georgetown's Policy

Georgetown Honor Council's Standards of Conduct Policy (2024) states:

"The question of how to acknowledge [AI-generated intellectual work], and whether it is to be allowed at all, is answered by individual course policies. It is, as always, the students' responsibility to be sure that they are following the rules laid out by their professors. Note that, as with all source material, this applies both to work taken directly from the AI generator and to work that has been paraphrased before being used in coursework. If you didn't generate the words yourself, say so by quoting and citing the source; if you generated the words but not the content and ideas, say so by citing the source."

Learning Resources

Prerequisites

- **Python experience.** We will use Python as the primary interface for Apache Spark (PySpark), data processing examples, cloud service SDKs, and some SQL interaction.
- **Familiarity with data concepts.** Basic understanding of data types, tabular data, and elementary statistics.

Tutorials to brush up on these skills:

- [git - the simple guide](#)

- [Learn git concepts, not commands](#) by Nico Riedmann
- [SQLBolt](#) - Learn SQL with simple, interactive exercises
- [The Missing Semester of Your CS Education](#) - Command line, version control, and developer tools

Books and Readings

There is no required textbook. A significant amount of material is available online, and I have selected specific chapters, articles, and papers from several sources. These will be provided via Canvas. I recommend completing assigned readings prior to lectures.

Software

You will use a variety of tools throughout the semester. We will discuss setup in the first few weeks.

Development Environment

- Visual Studio Code (recommended IDE)
- JetBrains tools (Alternative IDE, free academic accounts available)
- Git and GitHub Desktop
- Terminal/PowerShell and SSH

Databases and SQL

- DBeaver (database client)
- PostgreSQL (local SQL practice)

Programming

- Python (via Anaconda or standalone)
- Jupyter Notebooks
- PySpark

Cloud Platforms

- Snowflake (cloud data warehouse)
- Databricks (unified analytics platform)
- AWS (S3, Glue, EMR, SageMaker)

AWS Tools

- AWS CLI
- Boto3 (Python SDK)

Cloud Resources

You will use cloud resources from Snowflake, Databricks, and Amazon Web Services. We will set up accounts and environments during the first few weeks of class. Educational credits and trial accounts will

be used where available. My guess is that you will have some, but minimal, cloud service charges in this course.

Modules and Calendar

Class	Phase	Date	Title	Notes	Lab / Deliverable
1	Phase 1: Foundations	1/8/2026	Course Overview & Big Data Fundamentals	Overview of course structure, intro to big data concepts (3 Vs, policy applications), computing environment setup, version control fundamentals	GitHub accounts, "Hello World" GitHub Page, CLI basics
2	Foundations	1/15/2026	SQL 1: Tables & Data Modeling	History of SQL, relational model, data modeling basics (entities, relationships, normalization), foundational queries (SELECT, WHERE, JOIN)	DBeaver setup, Visual Paradigm, SQL Murder Mystery
3	Foundations	1/22/2026	SQL 2: Queries & Integration	Advanced queries (aggregation, subqueries, window functions), integrating SQL with Python (SQLite), alternative data formats (JSON, Parquet)	SQL Company Analysis. PROJECT 1: Country selection due
4	Foundations	1/29/2026	Cloud Fundamentals & Architecture	Scaling technologies (vertical vs. horizontal), cloud service models (IaaS/PaaS/SaaS), computer architecture concepts, metadata, intro to AWS CLI	AWS account setup, IAM, S3, permissions basics
—	Optional	TBD	Data Center Tour	Optional tour of a local datacenter	—
5	Phase 2: Cloud Platforms	2/5/2026	Cloud #1: Snowflake Introduction	Cloud data warehouse concepts, Snowflake architecture (virtual warehouses, storage separation), applying SQL in cloud context	Snowflake environment setup, first queries
6	Cloud Platforms	2/12/2026	Cloud Data Engineering	Data lakes vs. warehouses vs. lakehouses, data pipelines, ELT patterns, Snowflake data science features, introduction to cloud SDKs	Snowflake Notebooks, staging data. PROJECT 1: Outline due
7	Cloud Platforms	2/19/2026	AWS Data Services	S3 deep dive, Aurora, RDS, Redshift overview, Glue basics, Lake Formation, Iceberg and open table formats	AWS Glue job, S3 data organization
8	Cloud Platforms	2/26/2026	Data Ethics, Governance & Regulations	Data ethics frameworks, privacy laws (GDPR, CCPA, state laws), disaster recovery, cloud security basics, NIST and AI governance frameworks	PROJECT 1: Country Presentations

—	—	3/5/2026	NO CLASS – Spring Break	—	PROJECT 2: Topic selection due (3/11/2026)
9	Phase 3: Distributed Computing	3/12/2026	Distributed Computing Fundamentals	Why distributed? Hadoop/MapReduce to Spark evolution, disk vs. memory access, parallel algorithms, fault tolerance, CAP theorem basics	Local Spark installation and first job
10	Distributed Computing	3/19/2026	Cloud #2: Databricks Introduction	Databricks architecture, Delta Lake, unified analytics platform, notebooks environment, cluster management	Databricks workspace setup, notebook basics. PROJECT 2: Dataset identification due
11	Distributed Computing	3/26/2026	Spark Deep Dive	Spark SQL, DataFrames, transformations vs. actions, lazy evaluation, introduction to MLlib, streaming concepts and latency	Databricks Spark analysis
—	—	4/2/2026	NO CLASS – Easter Break	—	PROJECT 2: Outline/methodology due (4/8/2026)
12	Distributed Computing	4/9/2026	AWS Compute & ML Infrastructure	EC2 fundamentals, EMR for Spark at scale, SageMaker overview, GenAI/BI tool landscape, cost considerations and right-sizing	EMR cluster or SageMaker notebook
13	Phase 4: Advanced Models & Synthesis	4/16/2026	Alternative Data Paradigms	Vector databases (embeddings, semantic search, RAG applications), graph databases and Neo4j, when to choose non-relational models	Introduction to Cypher, vector similarity demo. PROJECT 2: Draft due (optional feedback)
14	Synthesis	4/23/2026	Big Data as Problem-Solving Paradigm	Architecture decision frameworks, evaluating trade-offs, emerging trends, course synthesis	PROJECT 2: Final Presentations
	First day of final examinations	5/1/2026			PROJECT 2: Final paper and any revisions due

Project Timeline Summary

Project 1: Country Data Landscape Presentation

Country selection: 1/22/2026 (Class 3) | Outline: 2/12/2026 (Class 6) | Presentation: 2/26/2026 (Class 8)

Project 2: Final Project

Topic selection: 3/8/2026 (End of Spring Break) | Dataset identification: 3/19/2026 (Class 10) | Outline/methodology: 4/5/2026 (End of Easter Break) | Draft (optional): 4/16/2026 (Class 13) | Final presentation: 4/23/2026 (Class 14) | Final paper and revisions due: 5/1/2026 (First day of exams)

Warning

IT IS YOUR RESPONSIBILITY TO MANAGE CLOUD CREDITS AND RESOURCES PROVIDED TO YOU. YOU MUST SHUT DOWN YOUR CLOUD RESOURCES WHEN NOT IN USE.

Learning Activities, Communication and Evaluation

This is a hands-on, practical, workshop style course that provides opportunities to use the tools and techniques discussed in class. Although this is not a programming course per se, there is programming involved.

Lectures and Labs

This course is split into a lecture/lab format, where every class session will have a lecture portion, and most sessions will have an in-class lab portion:

- During the lecture, we will discuss the concepts and techniques as well as the history and development of these big data tools and cloud platforms.
- During the lab sessions, you will be completing exercises and following examples which are designed to show you how to implement ideas and concepts with various tools. We will start the labs in class, but we might not finish. It is your responsibility to complete the labs (**which are part of the grade**).

Lectures may not cover all the material, and some topics will be introduced in the lab or through readings/assignments.

Office Hours

Instructors and TAs are available to meet to answer questions, review material, and support your learning.

Readings

On certain weeks, readings will be assigned to prepare you for the lecture material being presented. These readings should take an hour or less per week.

Online Quizzes

Quizzes will be given a few times during the semester during lab or lecture at random intervals and times. Quizzes ensure you are keeping up with the material presented in the class. Quizzes are meant to be brief and low-stress with a time limit of 5-10 minutes. The material will be drawn from lectures, labs, and readings.

Lab Deliverables

Each lab will have a deliverable. It is essential that you learn the skills presented in the labs so that you can effectively complete the assignments and the big data project. The lab deliverables can sometimes be completed during lab; however, it is your responsibility to complete the deliverable as part of your work outside of lecture/lab time.

Projects

There are two projects that are designed to enhance your learning of the data environment and how data is used throughout the world. I am a strong believer that projects and the hands-on work they involve are one of the best ways to learn this material. My hope is that at the end of the course you will have the beginnings of a portfolio that you can share with future colleagues and employers, as well as refer to for your future self.

Project 1: Global Data Landscape

The legal and physical environments for data vary significantly across the world and are constantly evolving. For this project, students will select a specific (non-U.S.) country to research and will analyze its data ecosystem, examining both technical infrastructure and regulatory frameworks.

- a) On the infrastructure side, students will investigate the presence and operations of major cloud providers, data center capacity and connectivity, internet access methods and penetration rates, and the country's readiness for advanced data-driven technologies.
- b) On the regulatory side, students will examine privacy and data protection laws, cross-border data transfer restrictions, cybersecurity requirements, and compliance frameworks that affect how organizations can collect, store, and process data.

The analysis should consider how historical context, regional influences, economic conditions, and cultural factors shape the country's approach to data governance, and how these regulatory choices interact with infrastructure investment and technology adoption.

The project culminates in two deliverables: a concise presentation highlighting key findings and a written briefing report providing deeper analysis. Students are encouraged to include relevant comparisons with peer nations to provide context.

Milestones:

Country selection: Thursday, January 22 (Class 3)

Outline and preliminary sources: Thursday, February 12 (Class 6)

Presentation: Thursday, February 26 (Class 8)

Final report: Wednesday, March 11

Students may work individually or in small teams.

Project 2: Big Data Analytics Project

The final project challenges students to demonstrate mastery of big data analytics applied to public policy. Working in teams of up to four, students will select a policy area of interest, identify relevant datasets, and conduct an initial analysis using cloud infrastructure and scalable techniques.

The emphasis is on methods, not dataset size. Students should apply the tools and approaches covered throughout the semester (cloud platforms, distributed computing, data pipelines, and appropriate analytical techniques) even when working with datasets that could technically fit on a single machine. The goal is to demonstrate fluency with scalable infrastructure and to produce a proof-of-concept analysis that lays the groundwork for future research or funding opportunities.

The framing for this project is a pitch to a grant funder or policy board. Teams will explain why their policy question matters, present their initial analysis and findings, and propose a path forward for larger-scale investigation.

Deliverables:

- Project website (e.g., GitHub Pages) showcasing the analysis, working code, and visualizations
- In-class presentation simulating a grant funding pitch
- Policy research paper (under 3,000 words) providing deeper context, methodology, and implications

Concise, directed writing is preferred over length. The paper should make a focused argument rather than fill space.

Milestones:

- Topic selection: March 8 (end of Spring Break)
- Dataset identification: March 19 (Class 10)
- Outline/methodology: April 5 (end of Easter Break)
- Draft for feedback (optional): April 16 (Class 13)
- Website and presentation: April 23 (Class 14)
- Final paper and project revisions: May 1 (first day of finals)

Milestones are graded for completion. Their purpose is to provide guidance and identify potential issues early. Schedule a meeting with me if you're stuck.

Teams: Students form their own teams. If you're having difficulty finding a team, please let me know and I'll help coordinate.

University Policies

Mental Health Resources

Mental health is a serious and growing problem across the world and in universities in particular. Georgetown offers counseling and other services. More information is available here:

<https://studenthealth.georgetown.edu/mental-health/>

Academic Resource Center (ARC)/Disability

If you believe you have a disability, contact the Academic Resource Center (arc@georgetown.edu) or visit the ARC website (<https://academicsupport.georgetown.edu/disability/>) for further information. The ARC is the Georgetown office responsible for reviewing documentation provided by students with disabilities and for determining reasonable accommodations in accordance with the Americans with Disabilities Act (ADA) and University policies. The ARC is located in the Leavey Center, Suite 338 (202-687-8354).

Important Academic Policies and Academic Integrity

McCourt School students are expected to uphold the academic policies set forth by Georgetown University and the Graduate School of Arts and Sciences. Graduate students should therefore familiarize themselves with all the rules, regulations, and procedures relevant to their pursuit of a Graduate School degree. The policies are located at:

<https://sites.google.com/a/georgetown.edu/gsas-graduate-bulletin/vi-academic-integrity-policies-procedures>.

Statement on Sexual Misconduct

Georgetown University and its faculty are committed to supporting survivors and those impacted by sexual misconduct, which includes sexual assault, sexual harassment, relationship violence, and stalking. Georgetown requires faculty members, unless otherwise designated as confidential, to report all disclosures of sexual misconduct to the University Title IX Coordinator or a Deputy Title IX Coordinator. If you disclose an incident of sexual misconduct to a professor in or outside of the classroom (with the exception of disclosures in papers), that faculty member must report the incident to the Title IX Coordinator, or Deputy Title IX Coordinator. The coordinator, will, in turn, reach out to the student to provide support, resources, and the option to meet. [Please note that the student is not required to meet with the Title IX coordinator.]. More information about reporting options and resources can be found on the Sexual Misconduct Website:

<https://sexualassault.georgetown.edu/resourcecenter>.

If you would prefer to speak to someone confidentially, Georgetown has a number of fully confidential professional resources that can provide support and assistance. These resources include:

- Health Education Services for Sexual Assault Response and Prevention: confidential email sarp@georgetown.edu
- Counseling and Psychiatric Services (CAPS): 202.687.6985

Policy on Student Observance of Religious Holidays

Georgetown University promotes respect for all religions. Any student who is unable to attend classes or to participate in any examination, presentation, or assignment on a given day because of the observance of a major religious holiday (see below) or related travel shall be excused and provided with the opportunity to make up, without unreasonable burden, any work that has been missed for this reason and shall not in any other way be penalized for the absence or rescheduled work. Students will remain responsible for all assigned work. Students should notify professors in writing at the beginning of the semester of religious observances that conflict with their classes. The Office of the Provost, in consultation with Campus Ministry and the Registrar, will publish, before classes begin for a given term, a list of major religious holidays likely to affect Georgetown students. The Provost and the Main Campus Executive Faculty encourage faculty to accommodate students whose bona fide religious observances in other ways impede normal participation in a course. **Students who cannot be accommodated should discuss the matter with an advising dean.**

The McCourt Policy-Writing Center (MPWC)

The McCourt Policy-Writing Center (MPWC) provides free policy-writing expertise exclusively to McCourt students to help them communicate effectively in the classroom and their careers. Our services for students include our monthly lunch series with expert guest speakers; [online modules](#); and [individual appointments](#). Look out for our weekly newsletters for upcoming events and email mpwc@georgetown.edu for general inquiries.